

HOME SOC LAB

Wazuh-Based Security Monitoring, Detection Engineering & Active Response — Project Report

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1. Executive Summary

This report documents the design, build, and validation of a home Security Operations Center (SOC) lab centered on Wazuh, an open-source SIEM/XDR platform. The project went beyond installing and configuring the platform: it covers enrolling and monitoring two heterogeneous endpoints (Windows and Linux), extending Wazuh's default detection coverage with two original correlation rules mapped to MITRE ATT&CK, building a custom monitoring dashboard, and configuring automated active response for brute-force containment.

Every capability built was validated against real activity rather than assumed to work from configuration alone. This included generating genuine SSH brute-force attempts, enabling a Windows Guest account, and creating/modifying/deleting monitored files, then confirming that each action produced the expected alert, and — in the case of active response — the expected containment and recovery behaviour.

2. Objectives

- Deploy a working Wazuh manager and enroll heterogeneous endpoints (Windows 10 and Ubuntu Linux) as monitored agents.
- Extend endpoint visibility with Sysmon on the Windows agent for richer process, file, and network telemetry.
- Generate and review real telemetry end-to-end before building detections on top of it.
- Build a consolidated SOC dashboard for day-to-day monitoring of authentication and account-change activity.
- Enable File Integrity Monitoring and author original detection rules beyond Wazuh's default ruleset, each mapped to a MITRE ATT&CK technique.
- Configure automated active response so that a qualifying detection results in automatic containment, not just an alert.
- Validate the full detect → respond → recover cycle against self-generated attack activity.

3. Lab Environment

The lab runs entirely inside Oracle VirtualBox on a Windows host, with all VMs sharing an isolated lab network.

Hypervisor	Oracle VirtualBox, hosted on a Windows PC (32 GB RAM)
SIEM / XDR Platform	Wazuh v4.14.7 — single-node manager (node01)
Lab Network	192.168.68.0/24 (isolated VirtualBox network segment)
Agent 001 — IZAAN-Windows	192.168.68.131 · Microsoft Windows 10 Pro, build 10.0.19045.3803 · 2 vCPU / 4 GB RAM · Sysmon installed
Agent 002 — IZAAN-Linux	192.168.68.130 · Ubuntu 24.04.5 LTS · SSH exposed for brute-force testing

Figure 3.1 (below) confirms both agents enrolled and active under the manager, with Wazuh reporting one Windows and one Ubuntu endpoint in the default group.

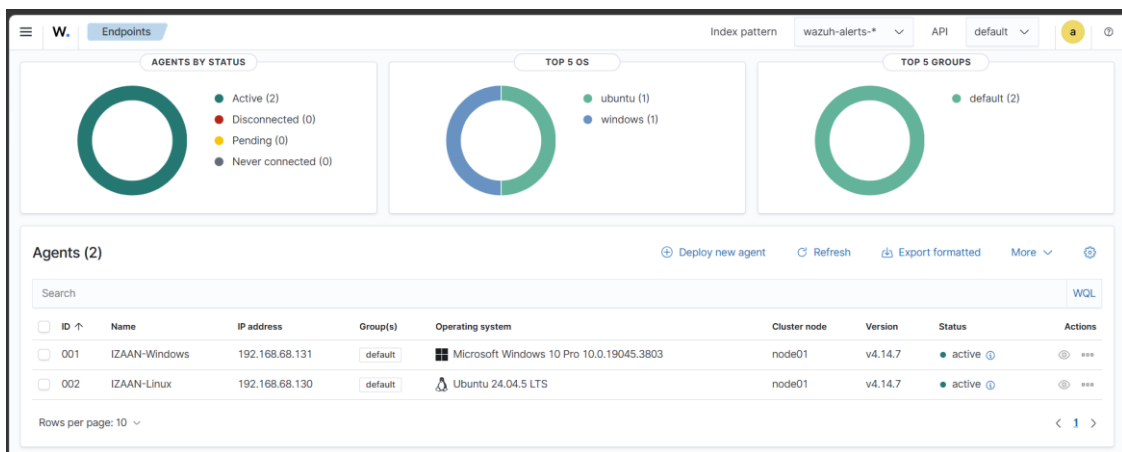


Figure 3.1 — Wazuh Endpoints overview: two agents enrolled and active (IZAAN-Windows, IZAAN-Linux).

4. Methodology

The build was carried out in six phases, moving from raw deployment through to a fully validated detection-and-response pipeline.

4.1 Phase 1 — SOC Server Build & Deployment

Deployed the Wazuh manager (v4.14.7) as the central component of the lab — responsible for log collection, indexing, correlation, and alerting for every endpoint enrolled afterward. This manager (node01) forms the single point of visibility used throughout the rest of the project.

4.2 Phase 2 — Agent & Sysmon Integration

Enrolled the Windows 10 and Ubuntu endpoints as Wazuh agents, then installed and configured Sysmon on the Windows agent to capture process creation, network connection, and file-related events beyond what native Windows Event Logs provide. The effect of this is visible directly in the endpoint's MITRE ATT&CK breakdown, which is populated almost entirely from Sysmon-derived detections.

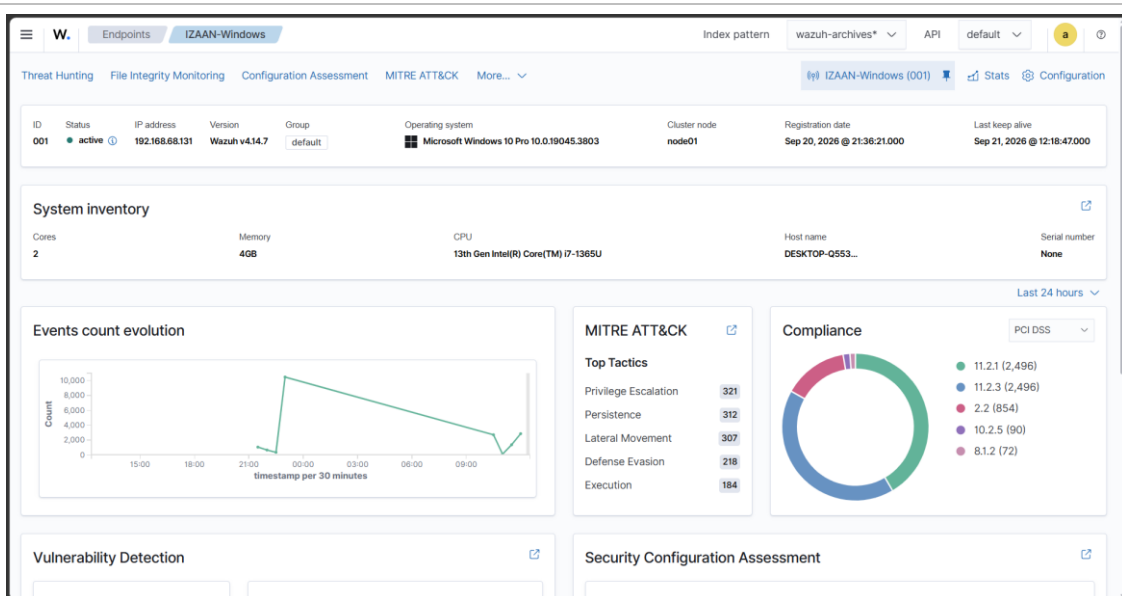


Figure 4.1 — IZAAN-Windows agent overview: system inventory, 24-hour event volume, and MITRE ATT&CK tactic breakdown (Privilege Escalation 321, Persistence 312, Lateral Movement 307, Defense Evasion 218, Execution 184).

4.3 Phase 3 — Telemetry Generation & Review

Before layering custom rules or dashboards on top of the pipeline, activity was generated on both endpoints and reviewed in Wazuh's Threat Hunting view to confirm telemetry was flowing correctly end-to-end from both agents. Over the review window, the Windows agent alone logged 22,028 total events, including 49 alerts at level 12 or above, 9 failed authentications, and 6 successful authentications — spread across alert groups spanning Sysmon, Windows security/application logs, the vulnerability detector, and SCA.

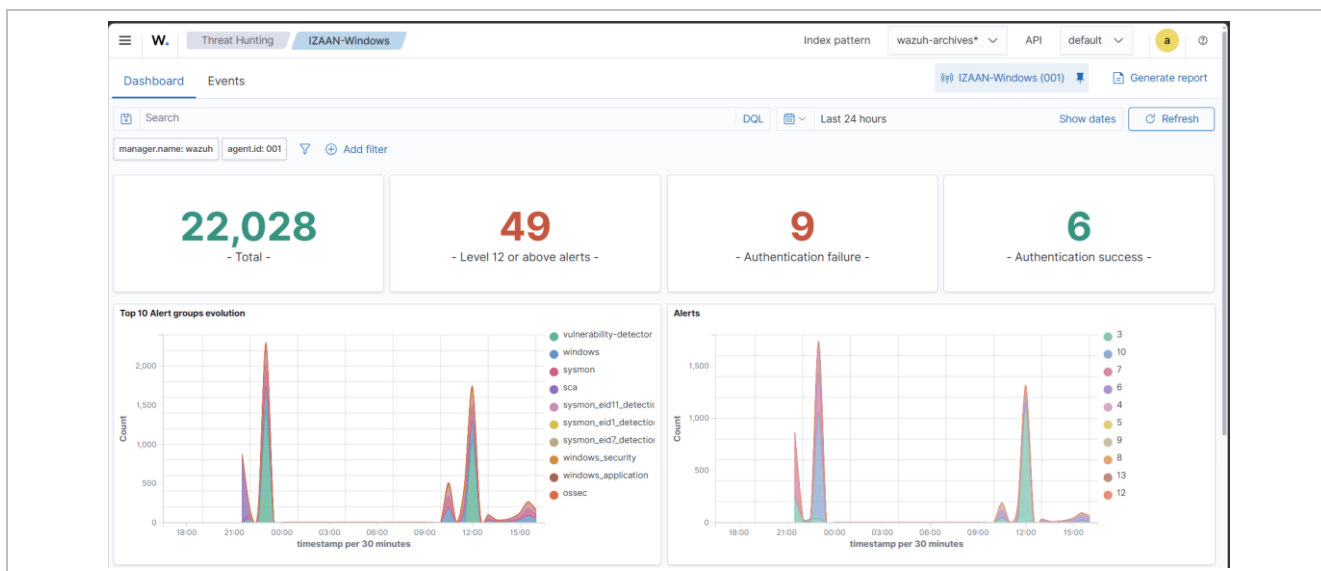


Figure 4.2 — Threat Hunting view for IZAAN-Windows: total event volume, high-severity alert count, and authentication activity over the review window.



Figure 4.3 — Same view, continued: top alert types, top rule groups, and top PCI DSS requirements touched by observed activity.

4.4 Phase 4 — SOC Dashboard Creation

Built a custom Wazuh dashboard to consolidate the signals most relevant to day-to-day monitoring into a single view: failed Windows logons, Linux SSH authentication activity by source, Windows account-change events over time, and a running total of user-account modifications. This turns raw log search into an at-a-glance operational view rather than requiring a fresh query for every check.

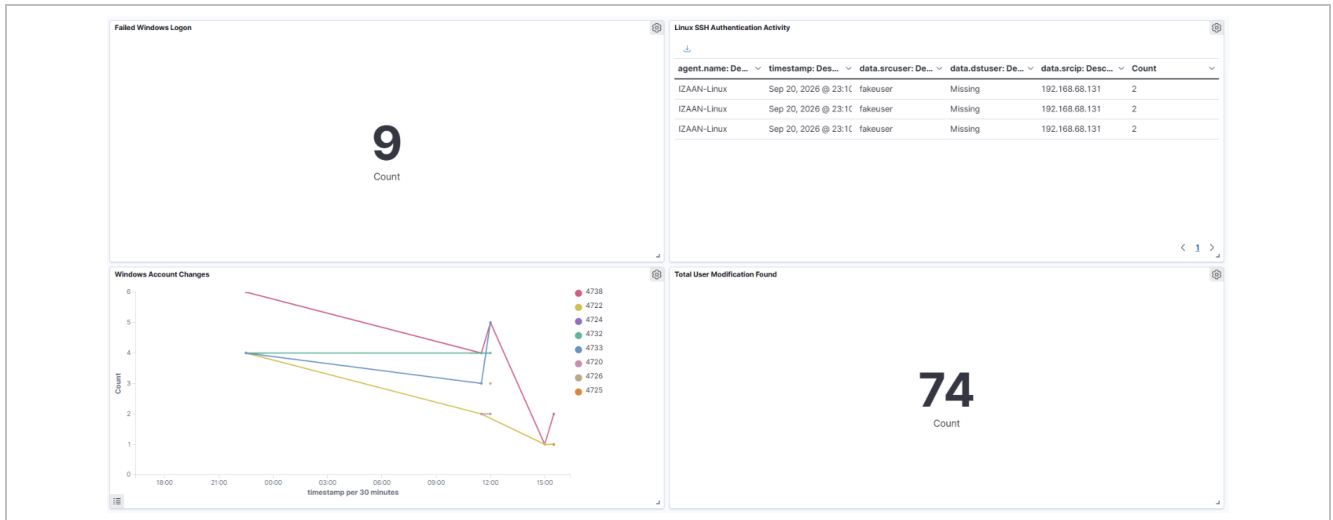


Figure 4.4 — Custom SOC dashboard: failed Windows logons (9), Linux SSH authentication attempts by source, Windows account-change trend, and total user modifications (74).

4.5 Phase 5 — File Integrity Monitoring & Custom Detection Rules

Enabled Wazuh's File Integrity Monitoring (syscheck) on a watched directory and validated it by creating, modifying, and deleting files — each action correctly classified as a distinct event type (added, modified, deleted).

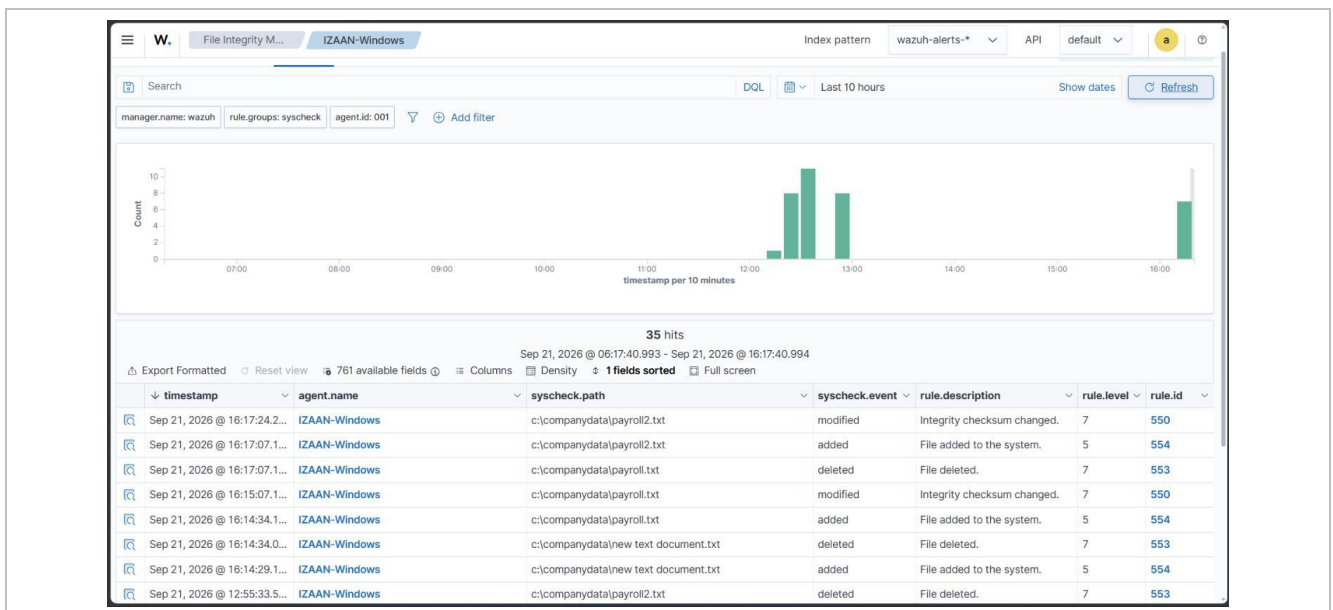


Figure 4.5 — File Integrity Monitoring events for IZAAN-Windows: 35 hits showing file additions, modifications, and deletions under a monitored directory, correctly classified by rule.

Went beyond Wazuh's default ruleset by authoring two original detection rules in local_rules.xml, each mapped to a specific MITRE ATT&CK technique:

- **Rule 100200 (level 12):** Fires when the built-in Windows Guest account is enabled (Event ID 4722, target user "Guest") — mapped to T1078 (Valid Accounts).
- **Rule 100101 (level 10, frequency 3 / timeframe 120s):** Fires on three or more failed SSH authentications from the same source IP within a 120-second window — mapped to T1110 (Brute Force).

```
< local_rules.xml
Ruleset Test Save

22 <group name="adduser, windows, windos_security, account_changed, ">
23
24 <rule id="100200" level="12">
25   <if_sid>60103</if_sid>
26   <field name="win.system.eventID">*4722</field>
27   <field name="win.eventdata.targetUserName">*Guest</field>
28
29   <description>Windows Guest account was enabled</description>
30
31   <mitre>
32     <id>T1078</id>
33   </mitre>
34
35   <group>
36     windows,
37     windows_account_management,
38     account_enabled,
39     guest_account,
40   </group>
41 </rule>
42
43 </group>
```

Figure 4.6 — Custom rule 100200 in local_rules.xml: detects the Windows Guest account being enabled, mapped to MITRE T1078.

Sep 21, 2026 @ 15:32:00.183 Windows Guest account was enabled

Expanded document [View surrounding documents](#) [View single document](#)

Table	JSON
r _index	wazuh-alerts-4.x-2026.09.21
r agent.id	001
r agent.ip	192.168.68.131
r agent.name	IZAAN-Windows
r data.win.eventdata.subjectDomainName	DESKTOP-Q553DD0
r data.win.eventdata.subjectLogonId	0xae1804
r data.win.eventdata.subjectUserName	Bob
r data.win.eventdata.subjectUserSid	S-1-5-21-652178336-3659581277-2001251109-1001
r data.win.eventdata.targetDomainName	DESKTOP-Q553DD0
r data.win.eventdata.targetSid	S-1-5-21-652178336-3659581277-2001251109-501
r data.win.eventdata.targetUserName	Guest
r data.win.system.channel	Security

Figure 4.7 — Rule 100200 firing in practice: alert generated when user “Bob” enabled the Guest account on IZAAN-Windows.

```
< local_rules.xml
Ruleset Test Save

44
45 <group name="local, syslog, sshd, authentication_failed, ">
46
47 <rule id="100101" level="10" frequency="3" timeframe="120">
48   <if_matched_sid>5760</if_matched_sid>
49
50   <same_source_ip />
51
52   <description>Multiple SSH login failures observed from the same source IP</description>
53
54   <mitre>
55     <id>T1110</id>
56   </mitre>
57
58   <group>authentication_failed, ssh_bruteforce, credential_access, </group>
59 </rule>
60
61 </group>
```

Figure 4.8 — Custom rule 100101 in local_rules.xml: detects repeated SSH login failures from a single source IP, mapped to MITRE T1110.

Time	data.parameters.alert.rule.description
Sep 21, 2026 @ 15:52:47.162	Multiple SSH login failures observed from the same source IP
Expanded document View surrounding documents View single document	
Table JSON	
f _index	wazuh-alerts-4.x-2026.09.21
f agent.id	002
f agent.ip	192.168.68.130
f agent.name	IZAAN-Linux
f data.command	add
f data.dstuser	izaan

Figure 4.9 — Rule 100101 firing in practice: alert generated after repeated failed SSH logins against IZAAN-Linux.

4.6 Phase 6 — Active Response Configuration

Configured Wazuh active response so that a qualifying brute-force alert automatically triggers a firewall-drop response, adding the offending source IP to iptables on the Linux agent — turning detection into containment without manual intervention.

Sep 21, 2026 @ 15:52:47.162	Host Blocked by firewall-drop Active Response
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Table JSON	
r _index	wazuh-alerts-4.x-2026.09.21
r agent.id	002
r agent.ip	192.168.68.130
r agent.name	IZAAN-Linux
r data.command	add
r data.dstuser	izaan
r data.origin.module	wazuh-execd
r data.origin.name	node01
r data.parameters.alert.agent.id	002
r data.parameters.alert.agent.ip	192.168.68.130
r data.parameters.alert.agent.name	IZAAN-Linux

Figure 4.10 — Active response event: the offending source IP added to iptables via the firewall-drop response after rule 100101 fired.

5. Attack Simulation & Validation

Rather than trusting configuration alone, every rule and response was validated against activity generated by hand. For the active response test specifically, the full cycle was walked through manually: repeated failed SSH logins were sent from the attacking host, and once Wazuh's active response blocked the source IP, that containment was confirmed independently — a ping to the target returned “connection timed out” and a further login attempt returned “permission denied.” The IP was then removed from iptables to restore legitimate access, closing the loop on detect → respond → recover.

Scenario	Trigger	Expected Detection	Observed Result
SSH brute force	Repeated failed SSH logins from one source IP	Rule 100101 fires (T1110)	Alert generated; active response blocked the source IP automatically
Containment check	Ping + login attempt against the blocked IP	Traffic dropped by firewall-drop	Ping timed out; login returned “permission denied”
Recovery	Manual removal of the IP from iptables	Access restored	Connectivity and login access returned to normal

Scenario	Trigger	Expected Detection	Observed Result
Guest account enabled	Guest account enabled on IZAAN-Windows	Rule 100200 fires (T1078)	Alert generated with full event context (source user, target account)
File tampering	Files created, modified, and deleted in a monitored path	Syscheck FIM rules fire per event type	All three event types (added / modified / deleted) correctly captured
Account-change volume	Windows user/group modification events over time	Tracked on the custom dashboard	74 total user modifications captured and charted

6. Skills & Tools Demonstrated

- **SIEM / XDR:** Wazuh manager deployment and administration; agent enrollment across Windows and Linux; Wazuh ruleset customization in XML; Wazuh active response configuration.
- **Endpoint Telemetry:** Sysmon deployment and tuning on a Windows endpoint to extend telemetry beyond native Windows Event Logs.
- **Detection Engineering:** Authoring original correlation rules with level, frequency, and timeframe tuning, mapped to MITRE ATT&CK techniques T1078 and T1110.
- **Network & Host Hardening:** Automated (active response) and manual iptables-based source-IP blocking and cleanup.
- **Monitoring & Reporting:** Custom dashboard design in Wazuh; File Integrity Monitoring; log-based threat hunting and triage.
- **Virtualization:** Multi-VM lab networking in VirtualBox across a SIEM manager and two monitored endpoints.

7. Conclusion & Next Steps

This project delivered a working, validated home SOC covering deployment, telemetry, detection engineering, monitoring, and automated response — with every stage confirmed against real activity rather than left as unverified configuration. The two custom rules and the active response pipeline in particular demonstrate detection engineering beyond a default install, mapped to concrete MITRE ATT&CK techniques.

Natural next steps to extend this lab further:

- Broaden attack simulation with a structured framework such as Atomic Red Team, to test coverage against a wider set of ATT&CK techniques.
- Add a SOAR layer (e.g. TheHive or Shuffle) to move from active response scripting toward full case management and orchestration.
- Extend network-level visibility with an IDS such as Suricata or Zeek alongside the existing host-based telemetry.
- Publish the write-up and rule source (local_rules.xml) to a public repository as a portfolio piece, and distill the outcomes above into quantified resume bullet points.